



NCCER Module 22105



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Utility Tractors





Section One

Utility Tractors and Their Equipment

Objective

1. Describe the different types of utility tractors and their equipment.
 - a. Explain the drive system used on utility tractors.
 - b. Identify and describe the instruments used on utility tractors.
 - c. Identify and describe the controls used on utility tractors.
 - d. Identify and describe the hitches and attachments used on utility tractors.



Performance Task

Under the supervision of an instructor, trainees should be able to do the following:

There are no Performance Tasks associated with this section.



1.0.0 Utility Tractors and Their Equipment

- Less than 100 hp to over 300 hp.
- Used for staging, finish grading, fence installation, and moving construction trailers.
- Diesel engines, mechanical three-point hitches, drawbars, and an independent mechanical **power takeoff (PTO)**.
- Options: ROPS operator cab, electro-hydraulic (EH) three-point hitch, EH PTO, and EH selectable control valves (SCVs).



(A) OVERHEAD VIEW



(B) SIDE VIEW

1.1.0 Drive Train Components (1 of 2)

Components on a typical John Deere™ utility tractor



1 - ENGINE
2 - AIR CLEANER
3 - FUEL TANK
4 - TURBOCHARGER
5 - COOLING SYSTEM
6 - CLUTCH

7 - TRANSMISSION
8 - DIFFERENTIAL
9 - REAR AXLE
10 - PTO SHAFT
11 - MFWD AXLE
18 - THREE POINT HITCH



12 - VARIABLE DISPLACEMENT PUMP
13 - HYDRAULIC FLUID FILTER
14 - ROCKSHAFT AND SCV CONTROLS

15 - REAR SCV
16 - MIDMOUNT SCV (IF EQUIPPED)
17 - ROCKSHAFT UNIT



1.1.0 Drive Train Components (2 of 2)

- A diesel, gasoline, or propane engine provides power for these tractors.
- Some tractors have a **turbocharger**, which compresses air for better combustion and results in more power delivered to the crankshaft.
- The crankshaft transmits torque to either a clutch that provides power to a mechanical transmission or to a hydraulic, or **hydrostatic**, transmission.
- The main power train drives the power takeoff (PTO) shaft and mechanical front-wheel drive axles through separate EH clutches.
- Power from the engine is coupled to a variable-rate hydraulic system that powers the EH clutches, the **rockshaft unit**, the SCV controls to the rear SCVs, the mid-mount SCVs if equipped, and the rockshaft unit for the three-point hitch.



1.1.1 Transmissions (1 of 2)

Sliding gear — Two or more internal parallel shafts.

Constant mesh — Collar (or shuttle) shift and synchromesh.

- In the collar shift version, the selected free-running gears are locked to their shafts by a sliding collar or shuttle.
- In synchromesh transmissions, operators can select gears without clashing by synchronizing devices in the transmission that match the speeds of mating gears before they engage.

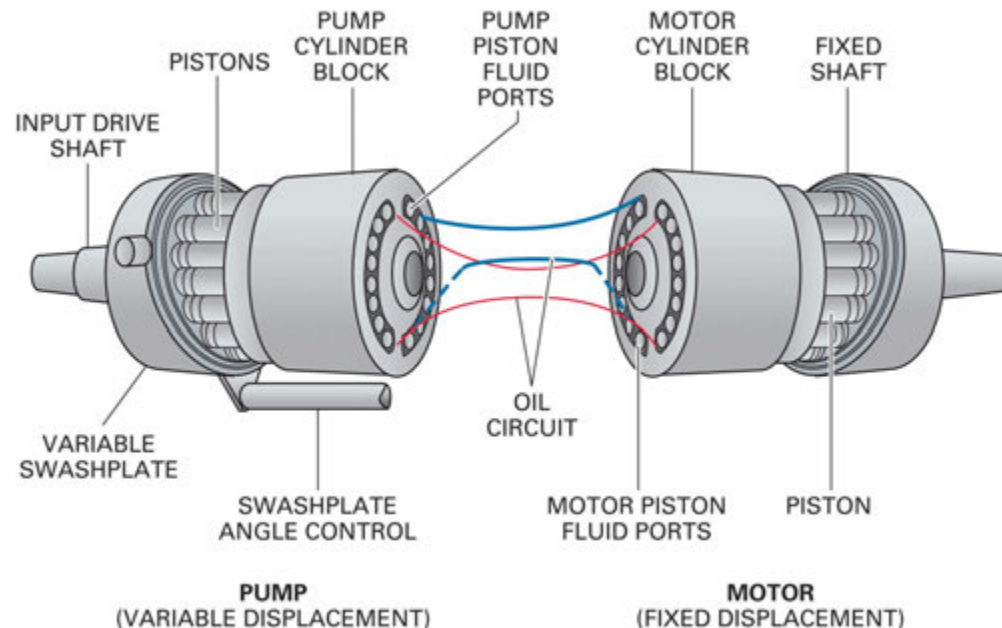
Hydraulic assist — A train of constantly-meshed gears that operators can shift while moving, without a clutch, to prevent interruption of the power flow through the transmission.



1.1.1 Transmissions (2 of 2)

Hydrostatic drive — A hydrostatic-drive transmission allows an infinite adjustment of tractor speed within a range without any gear shifting.

- The ratio of pressures between the supply and return ports transfers energy from the engine to the powered device.



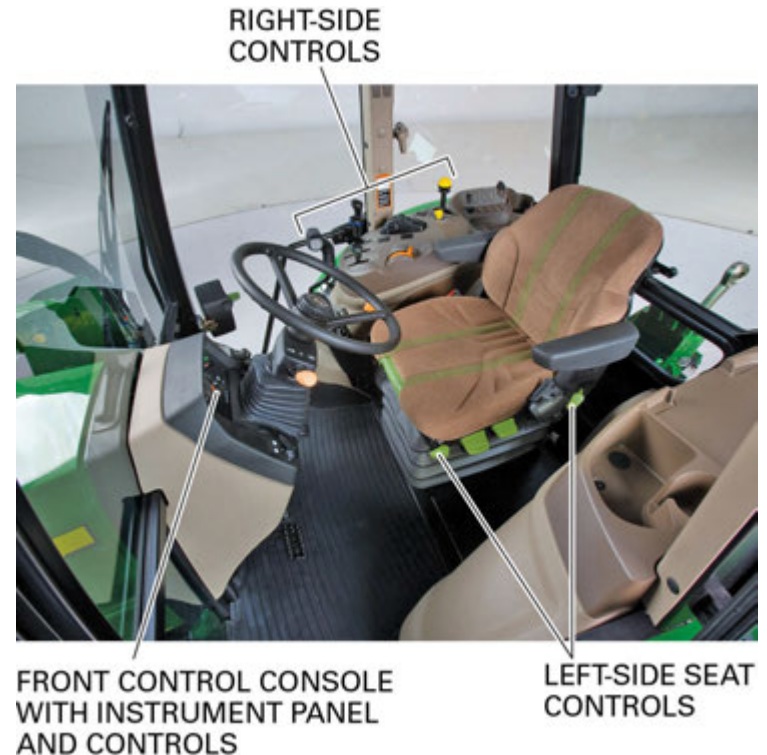
1.1.2 Clutches

- A tractor engine commonly uses a disk clutch to connect it to a mechanical transmission.
- A basic disk clutch consists of three main components: a drive plate, a clutch plate, and a pressure-plate assembly.
- Heavy-duty clutches contain more than one clutch disk **splined** to the output shaft and separated by steel plates that slide back and forth on the standoff bolts between the flywheel and pressure plate assembly.
- An overrunning clutch automatically engages in one direction but freewheels in the opposite direction using a number of roller bearings and ramps on the input shaft inside the output shaft.
- Cone clutches commonly control gear trains that supply auxiliary power, including PTOs.



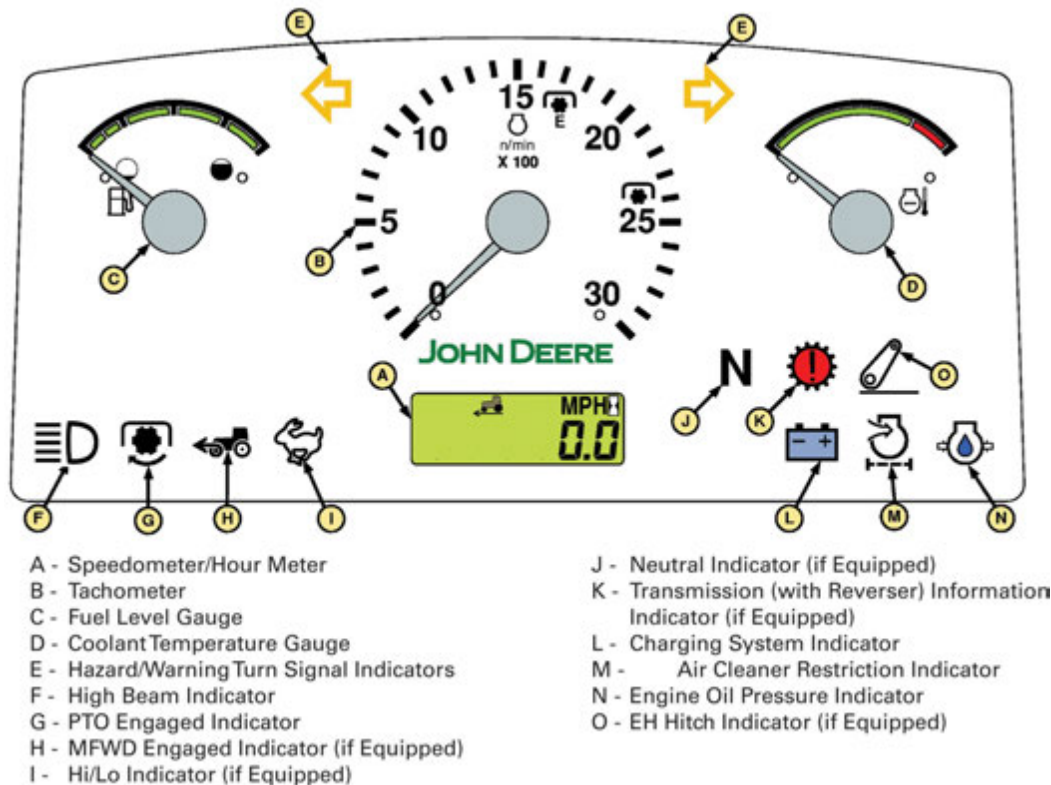
1.2.0 Instruments (1 of 2)

Study the operating manual to understand the specific instruments, including engine operating temperatures, charging system readings, oil and hydraulic system pressures, and other operating characteristics.



1.2.0 Instruments (2 of 2)

Operators must closely observe the tractor's instrument panel, located on the front control console, during operation to avoid serious damage.



1.2.1 Charging System Warning Indicator

- Indicates if the charging system is out of a normal range because the battery is discharging due to an excessive load or a malfunctioning alternator.
- Some tractors are equipped with an ammeter or voltmeter in addition to, or instead of, a warning indicator.



1.2.2 Engine Oil Pressure Warning Indicator

- Indicates if the engine oil pressure falls below an acceptable level.
- Some tractors are equipped with a warning indicator and/or an oil pressure gauge.
- Normal engine oil pressures vary widely due to the type of engine and manufacturer.



1.2.3 Fuel Level Gauge

- Indicates the amount of fuel in the tractor's fuel tank.
- On diesel-engine tractors, the gauge may contain a low fuel warning zone, or a low fuel warning light may be present.
- Avoid running out of fuel on diesel engine tractors because operators must bleed the fuel lines and injectors to remove air before restarting the engine.



1.2.4 Coolant Temperature Gauge

- Indicates the temperature of the coolant flowing through the engine.
- Operating temperatures vary between tractor makes and models.
- Most gauges indicate the normal operating range and needle near the center of the gauge.
- Warning lights may also display if the engine overheats.



1.2.5 Tachometer

- Indicates engine speed in revolutions per minute (rpm).
- Most tachometers are marked in hundreds on the meter face.
- The normal operating range varies between makes and models.
- Most tachometers also have special marks for operation of a PTO.



1.2.6 Speedometer/Hour Meter

- Indicates the forward speed of the tractor in miles per hour (mph) and the number of hours of operation.
- The speedometer is helpful when the effectiveness of an attachment depends on the speed of the tractor.
- Never operate PTO-driven equipment above the rated speed listed in the tractor's operator manual.



1.2.7 Additional Status and Warning Indicators

- Hazard warning/turn signal
- Headlight high beam
- PTO engaged
- MFWD engaged
- Hi/Lo lights
- Neutral lights
- Transmission (with reverser) information indicator
- Air cleaner restriction
- EH hitch

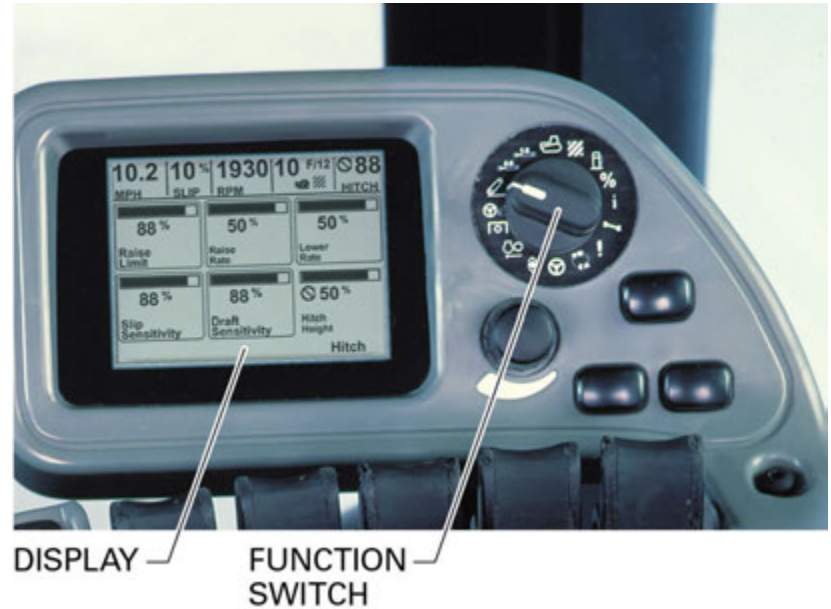
Some tractors may also have the following warning indicators that require terminating operation:

- Transmission wet-clutch low oil
- PTO clutch low oil



1.2.8 Performance Monitor

- These devices display digital values for ground speed, productivity, area covered, wheel slip, PTO speed, engine speed, distance, and hours since last service.
- Systems for agricultural use may also have global positioning system (GPS) guidance and attachment management systems.



1.3.0 Controls

Controls for most tractors are located on the seat, under the front control console, on the side and rear control console, and on the overhead air conditioning control console.

Some manufacturers color-code the controls. An example color code:

- **Yellow** — PTO or auxiliary power
- **Red** — Gear shift levers or tractor movement
- **Black** — Hydraulics or miscellaneous functions



1.3.1 Seat and Steering Wheel Adjustment

The operator should adjust the seat and steering wheel position and then fasten the seat belt before operating the tractor.



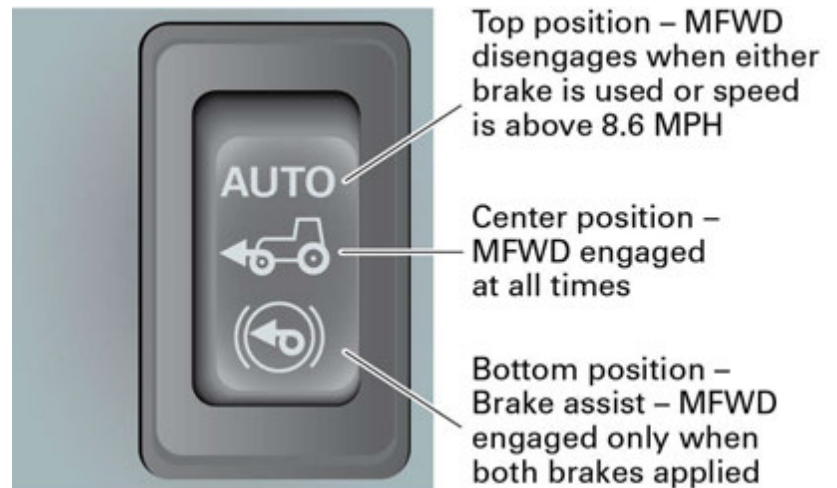
ADJUSTABLE SEAT

ADJUSTABLE
STEERING WHEEL

1.3.2 Transmission Shift Lever Operation

On tractors equipped with standard, synchronized, or hydraulic-assist mechanical transmissions, there are five basic transmission shift options:

- Range selector
- Gear selector
- High-Low shifter
- Reverser
- Front-wheel drive selector



1.3.3 Clutch Pedal Operation

- The clutch pedal engages and disengages power flow from the engine to a mechanical transmission.
- Tractors with a hydraulic assist transmission have an inching pedal in place of the clutch pedal.
- On tractors equipped with full power shift or a hydrostatic transmission, no clutch or inching pedal is present.



1.3.4 Brake Pedal Operation

- Tractors with two-wheel drive, including MFWD tractors, have two brake pedals on the right side of the operator platform.
- Full-time, four-wheel drive tractors have only one brake pedal.



1.3.5 Differential Lock Operation

- Some tractors have an electro-hydraulic differential lock that automatically engages when the rear wheels begin to slip, and disengages when the traction equalizes, a brake is applied, or a turn is attempted.
- Operators must not turn tractors with mechanical locks while the lock is engaged.
- To use a mechanical lock, the tractor wheels must be stopped or turning at the same speed before engaging the lock.



1.3.6 Tractor Light Operation

- Tractors are required to be equipped with legally suitable lights for any on-road use.
- Many tractors have front and rear work lights for off-road use.
- Most light switches have a number of positions for various operational conditions, including Off, Warning, Work Light, Transport 1, and Transport 2.



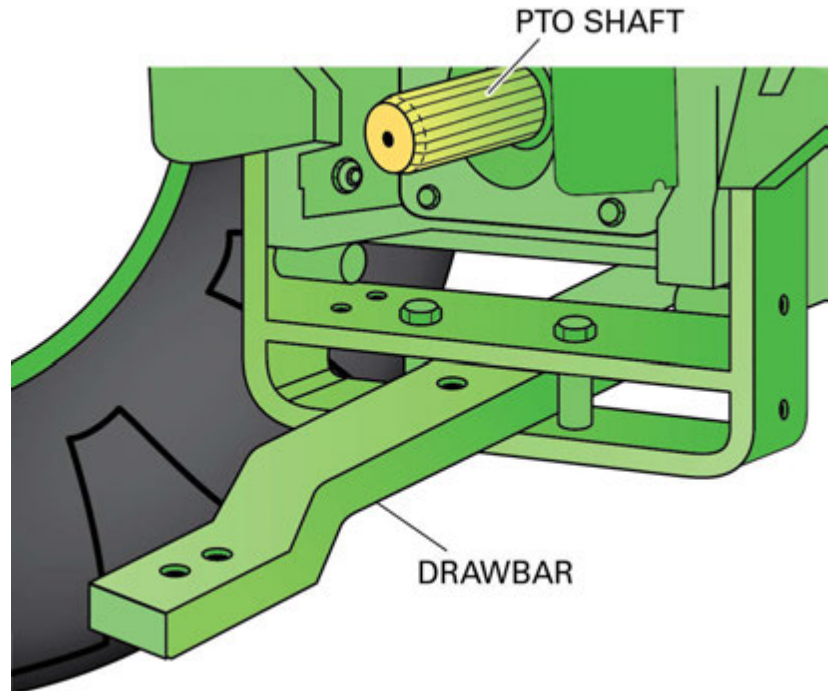
1.3.7 Cab Heating and Air Conditioning Operation

- Tractors with an optional enclosed operator's cab usually have a heating and air conditioning unit as standard equipment in the cab.
- Keeping the cab pressurized and the blower speed on high will keep dust out of the cab.



1.3.8 Power Takeoff Operation (1 of 4)

- A PTO (power takeoff operation) supplies mechanical power to certain attachments connected to the rear of the tractor at the three-point hitch or drawbar hitch.
- The PTO is a splined shaft at the rear of the tractor above the drawbar hitch



1.3.8 Power Takeoff Operation (2 of 4)

Tractors may have one of the following types of PTOs:

- **Transmission-driven** — When engaging a PTO selector lever, the tractor driveline transmission clutch operates the transmission and PTO at the same time.
- **Continuous-running** — A double-acting tractor driveline transmission clutch controls this type of PTO. With the clutch fully depressed, both the driveline and the PTO are disengaged. Only the PTO engages with the clutch released halfway; both the driveline and PTO engage with the clutch fully released.
- **Independent** — Most tractors use either a mechanical or an electro-hydraulic clutch PTO that operates separately from the driveline transmission and clutch.



1.3.8 Power Takeoff Operation (3 of 4)

- Tractors with the optional EH PTO can be converted to allow both 540 rpm and 1,000 rpm operation.
- Align the snap ring with the flats on the stub shaft, remove the snap ring, and pull the shaft out of the PTO housing.
- Insert the stub shaft into the PTO housing with the desired set of splines exposed.



PTO CONVERSION SHAFT

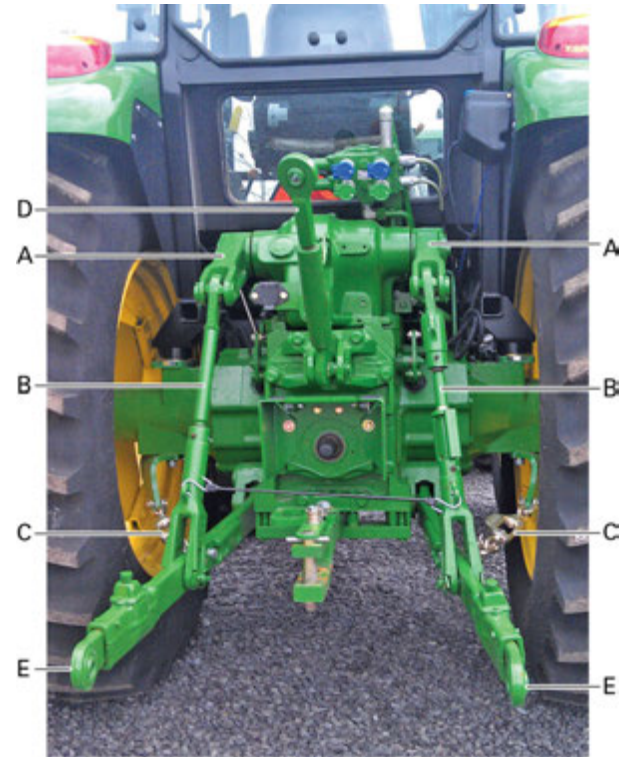
1.3.8 Power Takeoff Operation (4 of 4)

- Align the snap ring groove on the shaft with the snap ring groove in the PTO housing. Reinstall the snap ring over the flats on the shaft into both snap ring grooves.
- Make sure the PTO-driven attachment is properly installed to the tractor drawbar or three-point hitch and the PTO shaft of the attachment is properly connected to the splined shaft on the tractor with all shields in place.
- Set the PTO controls.



1.3.9 Hitch Control Operation (1 of 2)

- A three-point hitch can pull as well as raise or lower certain types of attachments.
- In the image, the attachment is connected to the **draft** links, which apply the pulling power of the tractor and establish the attachment's operating height above or below ground level (height/depth or draft).
- The tractor's engine must be running for the hitch controls to work.



1.3.9 Hitch Control Operation (2 of 2)

- **External raise/lower switches** — The rockshaft raises or lowers the hitch at a very slow rate. The switches are not functional when the hitch control lever in the cab is in the transport lock position.
- **Height limit control** — Changes the maximum hitch height.
- **Depth stop wheel** — Sets a mechanical stop at the desired working height (depth) of the attachment.
- **Rate-of-drop control** — Changes the rate of drop of the hitch.
- **Transport lock position** — Raises the hitch to its maximum height limit control setting and disables all raise/lower switches.
- **Load/height (depth) control** — Adjusts the draft response to raise and lower the attachment.



1.3.10 Selectable Control Valve (SCV) Operation (1 of 2)

- Most newer tractors are equipped with one or more SCVs to operate attachments with hydraulic motors and single- or double-action hydraulic cylinders.
- Tractors may have multiple sets of SCV couplers, depending upon the number of functions needed on the attachments they are pulling.

SCV COUPLER TO EXTEND
ATTACHMENT CYLINDER(S)



SCV COUPLER TO RETRACT
ATTACHMENT CYLINDER(S)

1.3.10 Selectable Control Valve (SCV) Operation (2 of 2)

Control levers usually have four operating positions:

- **Neutral** — Hydraulic oil pressure equalizes for both couplers of the SCV, which hydraulically locks a cylinder in a fixed position.
- **Extend** — Hydraulic oil is directed out of the top extend coupler of the SCV to the hydraulic cylinder at a rate that is proportional to the distance the lever is moved.
- **Retract** — Hydraulic oil is directed out of the bottom retract coupler of the SCV to a double-acting hydraulic cylinder at a rate that is proportional to the distance the lever is moved.
- **Float detent** — Both couplers are depressurized and the hydraulic cylinder operates in a float mode that allows the cylinder to extend and retract freely.



1.3.11 Ignition Switch Operation

The multipurpose ignition switch normally has four positions:

- **Accessory** — Powers devices such as radios that may be connected to accessory power.
- **Stop** — Stops the low-pressure primer fuel pump that feeds the mechanically driven high-pressure injector pump, closes the electrical fuel valve that stops the engine, shuts off all electrical power in the tractor.
- **Run** — Starts the low-pressure primer fuel pump, opens the electrical fuel valve that enables starting and running the engine, and initiates an instrument panel indicator bulb check.
- **Start** — Starts the engine.



1.4.0 Hitches and Attachments

- Attachments for the tractor normally connect at the rear.
- The tractor can pull the attachment with either a drawbar or carry it using a three-point hitch.
- Sometimes special tools or connectors are required to connect an attachment.



1.4.1 Types of Hitches and Attachments

- Finish grading can be accomplished with the adjustable rear grader blade.
- Box scrapers can level, grade, remove material, and backfill.
- The rotary cutter, often called a bush hog, operates from the tractor's PTO.
- When a tractor is equipped with a front-end loader bucket, a counterweight can be attached to the three-point hitch.



(A) TRACTOR WITH BOX BLADE ON REAR



(B) BASIC GRADER BLADE FOR TRACTOR



(C) ROTARY CUTTER (BUSH HOG)

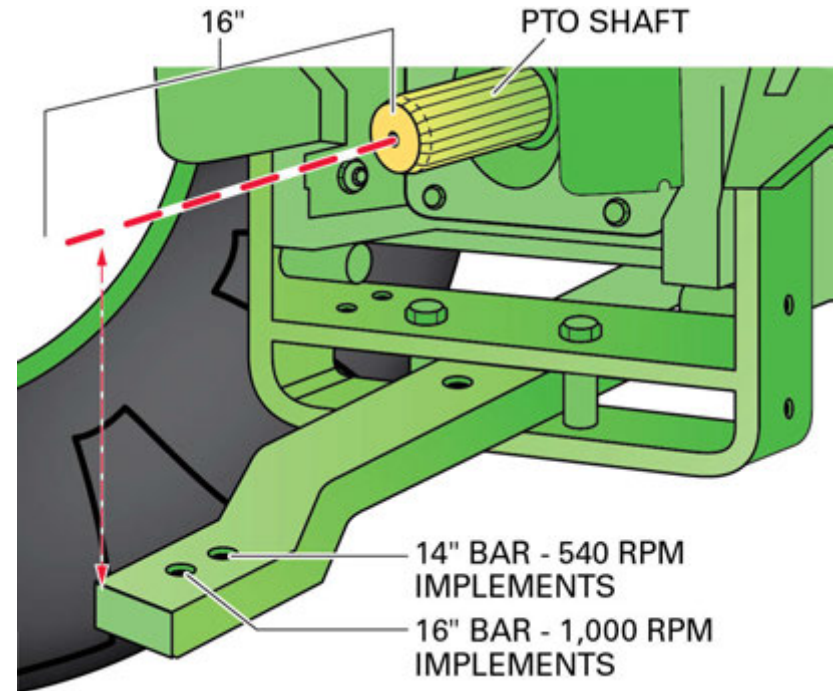


(D) FRONT BUCKET AND COUNTERWEIGHT

1.4.2 Drawbar Adjustments (1 of 3)

Length adjustment:

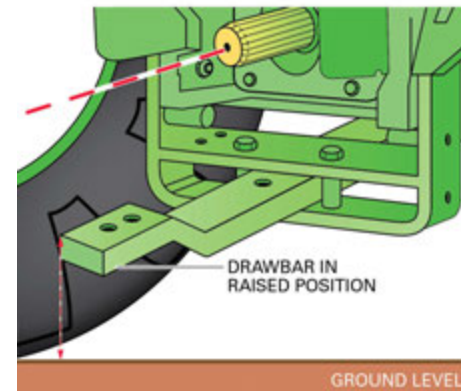
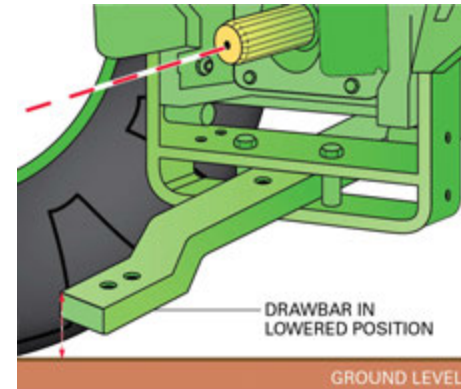
- The drawbar can be used in the short position or the extended position.
- The 14-inch short position is for PTO equipment rotating at a lower rpm.
- The 16-inch extended position usually applies to PTO equipment rotating at 1,000 rpm.
- In both positions, center the drawbar with the PTO shaft.



1.4.2 Drawbar Adjustments (2 of 3)

Height adjustment:

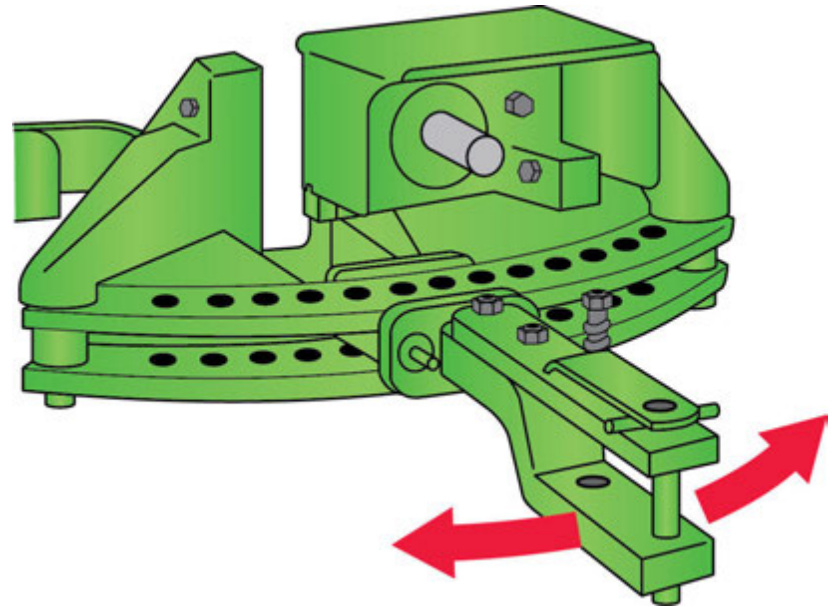
- Maintain the proper drawbar height and length to prevent the tractor from upsetting and to protect the PTO shaft.
- Adjust the height by inverting the drawbar.



1.4.2 Drawbar Adjustments (3 of 3)

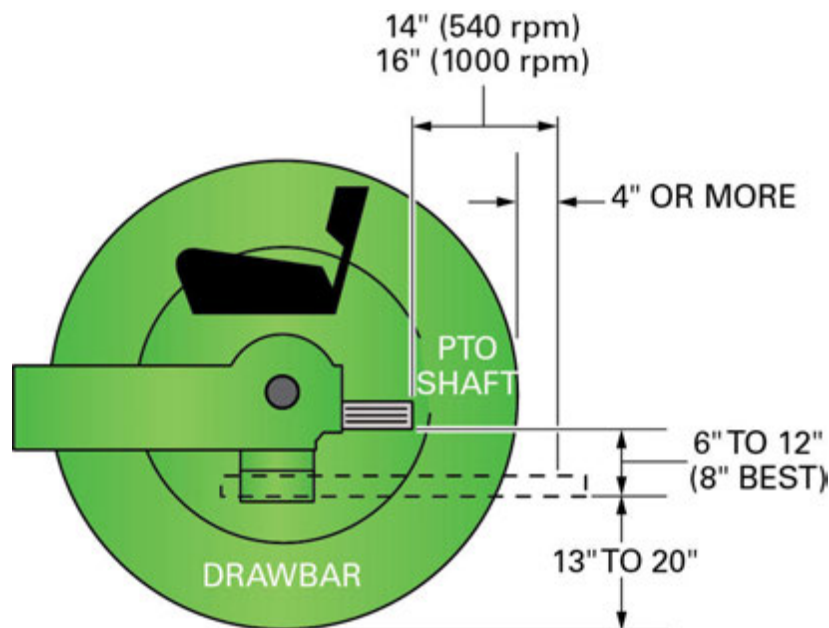
Lateral adjustment:

- Pinning the hitch in one of the swinging positions is for offset operations.
- Use the center fixed position for transporting.



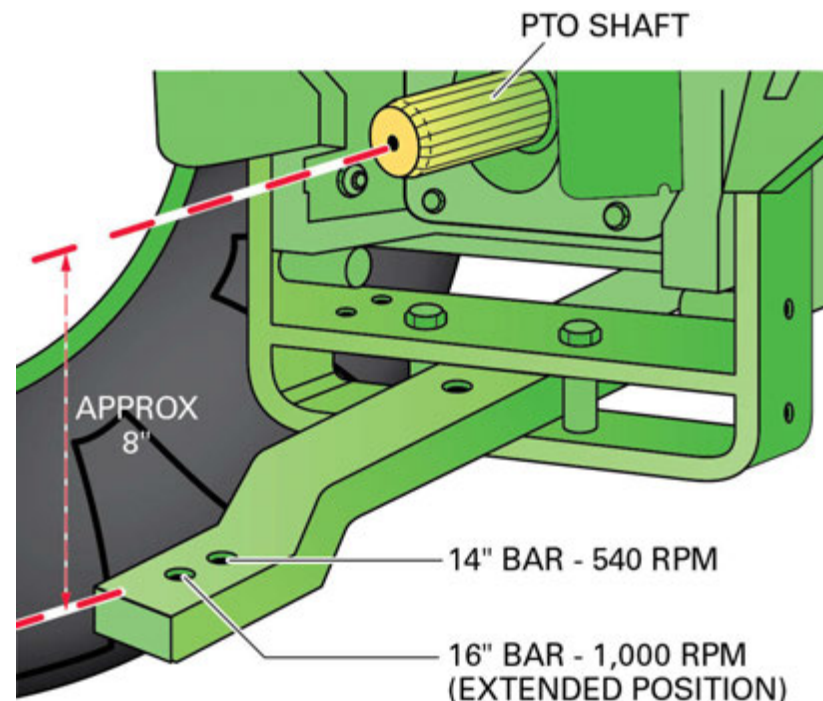
1.4.3 Connecting an Attachment to the Drawbar (1 of 2)

- Some attachments have a tongue for pulling, which must be connected to the tractor's drawbar.
- The drawbar height can be adjusted from 13 to 20 inches (33–51 cm) clearance between ground level and the drawbar hitch point.
- Be aware of hazards from lifting, cutting, pinching, and crushing.



1.4.3 Connecting an Attachment to the Drawbar (2 of 2)

- If the hitch will connect to an attachment that must be offset, extend the hitch.
- If the hitch connects to a PTO attachment, center the drawbar hitch point directly under PTO shaft and fasten it securely.
- Back the tractor into position so that the hole in the drawbar is in line with the hole in the attachment hitch.
- Set the hitch pin in place.



1.4.4 Ballasting a Tractor

- Some tractors may require wheel weights to prevent wheel slip when towing heavy loads.
- MFWD tractors usually require front end weights or front wheel weights, as well as rear tire weights to increase traction.
- For heavy attachments carried on a three-point hitch, you may have to add weights to the front of the tractor to act as counterweights for the attachment.
- To see where and how to install ballasting weights on the tractor being used, refer to its operator manual.
- Make sure to securely fasten the weights.



Wrap Up – Trade Terms (1 of 2)

Draft: In the heavy construction equipment trade, refers to the load-pulling capacity of a vehicle.

Hydrostatic: Relating to liquids at rest, or to the pressures they transmit, as in hydraulically powered controls or transmissions that stay in a neutral condition until the hydraulic fluid in them is put into motion or pressurized.

Power takeoff (PTO): Rotating mechanical output device on an engine or transmission to which cables, belts, or shafts can be connected to drive attachments or tools.

Rockshaft unit: Controls for equipment mounted on a three-point hitch.



Wrap Up – Trade Terms (2 of 2)

Splines: Parallel grooves running lengthwise on a shaft that engage matching grooves on a part that slides onto the shaft to prevent slippage.

Travel: With reference to heavy construction equipment, moving equipment from place to place. Can be either on the job site or over the public highway system.

Turbocharger: Exhaust-driven centrifugal air compressor used to increase air flow to an engine, which improves combustion efficiency and increases power.



1.0.0 Section Review Question 1

1. A type of transmission that permits infinite adjustment of speed within a given range without shifting is the _____.
- a. sliding gear
 - b. constant mesh
 - c. hydraulic assist
 - d. hydrostatic drive



1.0.0 Section Review Question 1 Answer

1. A type of transmission that permits infinite adjustment of speed within a given range without shifting is the _____.
- a. sliding gear
 - b. constant mesh
 - c. hydraulic assist
 - d. hydrostatic drive**



1.0.0 Section Review Question 2

2. The instrument that should be closely monitored when operating a tractor under heavy load is the _____.

- a. fuel level gauge
- b. coolant temperature gauge
- c. tachometer
- d. charging system indicator/gauge



1.0.0 Section Review Question 2 Answer

2. The instrument that should be closely monitored when operating a tractor under heavy load is the _____.

- a. fuel level gauge
- b. coolant temperature gauge**
- c. tachometer
- d. charging system indicator/gauge



1.0.0 Section Review Question 3

3. Which tractor control determines the desired ground speed of travel?
- a. Hitch control
 - b. Global positioning system
 - c. Transmission shift lever
 - d. Differential lock



1.0.0 Section Review Question 3 Answer

3. Which tractor control determines the desired ground speed of travel?
- a. Hitch control
 - b. Global positioning system
 - c. Transmission shift lever**
 - d. Differential lock



1.0.0 Section Review Question 4

4. Which of the following statements about tractor hitches and attachments is true?
- a. Front buckets are usually attached by a drawbar.
 - b. Bush hogs are powered by a PTO.
 - c. When equipped with a front-end loader bucket, a counterweight usually is attached to the front bumper.
 - d. All manufacturers use the same methods for attaching equipment according to consensus standards.



1.0.0 Section Review Question 4 Answer

4. Which of the following statements about tractor hitches and attachments is true?
- a. Front buckets are usually attached by a drawbar.
 - b. Bush hogs are powered by a PTO.**
 - c. When equipped with a front-end loader bucket, a counterweight usually is attached to the front bumper.
 - d. All manufacturers use the same methods for attaching equipment according to consensus standards.



Next...

Section 2.0.0
Starting, Operating, and Shutting
Down a Utility Tractor



*Read Sections 2.0.0 through 2.4.5 and
complete the Section Review.*